

Q.1) Read the current directory and display the name of the files, no of files in current directory

[10 Marks]

```
#include <stdio.h>

#include <stdlib.h>

#include <dirent.h>

int main() {

    DIR *dir = opendir(".");

    if (!dir) {

        perror("opendir");

        return 1;

    }

    struct dirent *entry;

    int count = 0;

    printf("Files in current directory:\n");

    while ((entry = readdir(dir)) != NULL) {

        if (entry->d_type == DT_REG) { // Only regular files

            printf("%s\n", entry->d_name);

            count++;

        }

    }

    printf("Total number of files: %d\n", count);

    closedir(dir);

    return 0;

}
```

Q.2) Write a C program which receives file names as command line arguments and display those filenames in ascending order according to their sizes. I) (e.g \$ a.out a.txt b.txt c.txt, ...)

[20 Marks ]

```
#include <stdio.h>

#include <stdlib.h>

#include <sys/stat.h>

#include <string.h>


int compare_sizes(const void *a, const void *b) {

    char *file1 = *(char **)a;

    char *file2 = *(char **)b;


    struct stat st1, st2;

    if (stat(file1, &st1) == -1) {

        perror(file1);

        exit(1);

    }

    if (stat(file2, &st2) == -1) {

        perror(file2);

        exit(1);

    }


    return (st1.st_size - st2.st_size);

}


int main(int argc, char *argv[]) {

    if (argc < 2) {

        printf("Usage: %s <file1> <file2> ...\n", argv[0]);

        return 1;

    }


    // Sort filenames by size using qsort

    qsort(argv + 1, argc - 1, sizeof(char *), compare_sizes);
```

```
printf("Files in ascending order of size:\n");  
for (int i = 1; i < argc; i++) {  
    struct stat st;  
    stat(argv[i], &st);  
    printf("%s (%ld bytes)\n", argv[i], st.st_size);  
}  
  
return 0;  
}
```